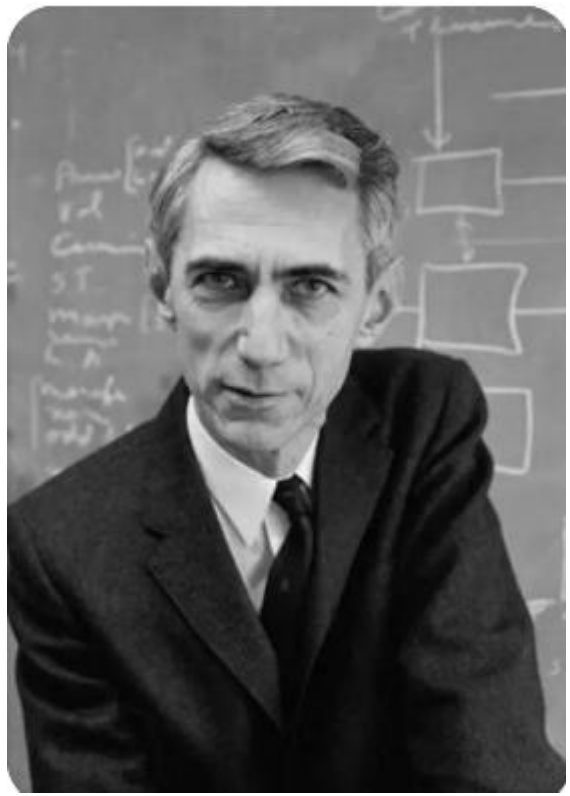
INFORMATION THEORY & CODING



Lecture 5

Review Summary

- **AEP Theorem** "*Almost all events are almost equally surprising.*" Specifically, if $X_1, X_2, \ldots$ are i.i.d. $\sim p(x)$, then

$$-\frac{1}{n}\log p(X_1, X_2, \ldots, X_n) \to H(X) \text{ in probability.}$$

- **Definition** The *typical set* $A_\epsilon^{(n)}$ is the set of sequences $x_1, x_2, \ldots, x_n$ satisfying

$$2^{-n(H(X)+\epsilon)} \leq p(x_1, x_2, \ldots, x_n) \leq 2^{-n(H(X)-\epsilon)}.$$

Review Summary

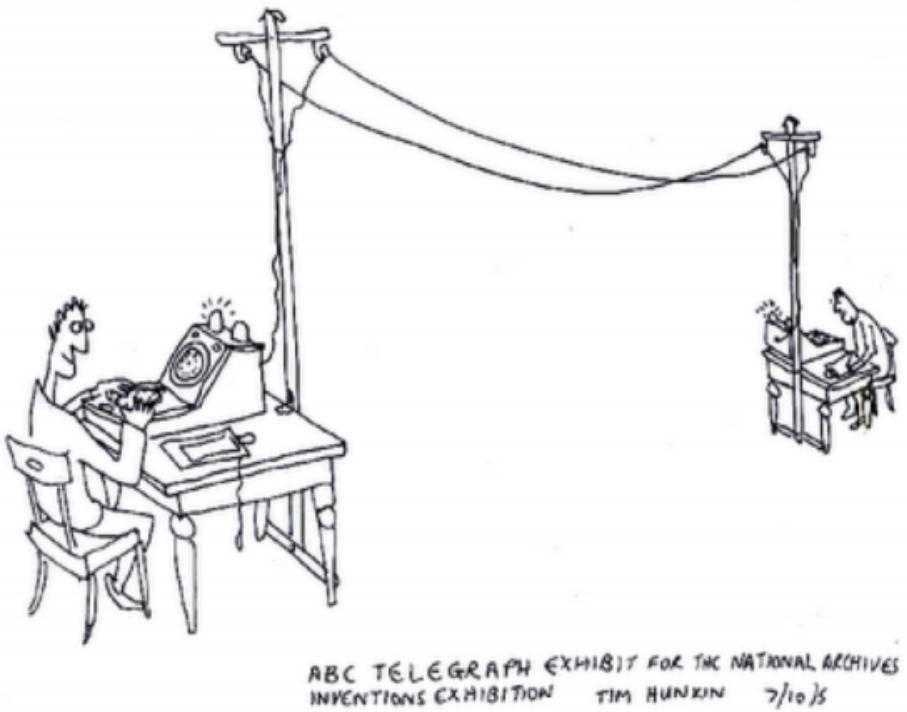
- **Properties of the typical set**
  1. If $(x_1, x_2, \ldots, x_n) \in A_\epsilon^{(n)}$, then $H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, x_2, \ldots, x_n) \leq H(X) + \epsilon$.
  2. $\Pr[A_\epsilon^{(n)}] > 1 - \epsilon$ for n sufficiently large.
  3. $|A_\epsilon^{(n)}| \leq 2^{n(H(X)+\epsilon)}$, where $|A|$ denotes the cardinality of the set A.
  4. $|A_\epsilon^{(n)}| \geq (1-\epsilon)2^{n(H(X)-\epsilon)}$ for n sufficiently large.
- **Theorem** Let X^n be i.i.d. $\sim p(x)$. There exists a code that one-to-one maps sequences x^n of length n into binary strings with

$$E\left[\frac{1}{n}\ell(X^n)\right] \leq H(X) + \epsilon$$

for n sufficiently large.

Source Coding

Which horse won in the horse racing?



Source Coding

Which horse won in the horse racing?

X	Pr	Code I	Code II
0	1/2	000	0
1	1/4	001	10
2	1/8	010	110
3	1/16	011	1110
4	1/64	100	111100
5	1/64	101	111101
6	1/64	110	111110
7	1/64	111	111111

$$H(X) = -\sum p_i \log p_i = 2 \text{ bits}$$

Which code is better?

Source Coding (Data Compression)

- We interpret that $H(X)$ is the best achievable data compression.

- We want to develop practical lossless coding algorithms that approach, or achieve the entropy limit $H(X)$.

Source Coding

- **Notation.** The *concatenation* of two strings $\underline{x}$ and $\underline{y}$ is denoted by $\underline{xy}$. The set of all strings over a finite alphabet $\mathcal{D}$ is denoted by $\mathcal{D}^*$.

- **Definition.** A *source code* C for a random variable X is a map

$$C : \quad \mathcal{X} \to \mathcal{D}^*$$
$$x \mapsto C(x)$$

where $C(x)$ is the codeword associated with x. Let $\ell(x)$ denote the length of $C(x)$.

Source Coding

- **Definition.** The *expected length* $L(C)$ of a source code $C(x)$ for a random variable X with probability mass function $p(x)$ is

$$L(X) = E\ell(X) = \sum_{x \in \mathcal{X}} p(x)\ell(x).$$

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$L_1(X) = ?$

$L_2(X) = ?$

Source Coding Applications

- Magnetic recording: cassette, hardrive, USB ...
- Speech compression
- Compact disk (CD)
- Image compression: JPEG

Still an active area of research:

- Solid state hard drive
- Sensor network: distributed source coding

Source Coding: A few examples

Assume $\mathcal{D} = \{0, 1, 2, \ldots, D-1\}$ in general (but often $D = 2$).

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$\mathcal{X} = \{1, 2, 3, 4\}$ and
$\mathcal{D} = \{0, 1\}$

x	p(x)	$c(x)$
1	1/2	0
2	1/4	10
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  - It's easy to decode:

$$0101101111111110100 \rightarrow 0, 10, 110, 111, 111, 110, 10, 0$$
$$1, 2, 3, 4, 4, 3, 2, 1$$

Source Coding: A few examples

- Example 2

$\mathcal{X} = \{1, 2, 3\}$ and
$\mathcal{D} = \{0, 1\}$

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Source Coding: A few examples

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  - $H(X) = 1.58$ bits, $L(C) = E\ell(X) = 1.66 > H$ bits
  - Can we easily decode?

$$10110010 \rightarrow 2, 3, 1, 1, 2$$

Source Coding: Set of codes

- For $\mathcal{X} = \{1, 2, 3, 4\}$ and $\mathcal{D} = \{0, 1\}$, consider

x	$p(x)$	C_I	C_{II}	C_{III}	C_{IV}
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  - Which code is best? Would we prefer C_I or C_{II}?

    Consider C_I and decode string: 00001. It would come from $1, 2, 1, 2, 3$ or $2, 1, 2, 1, 3$ or $1, 1, 1, 1, 3$, or etc.

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  - Which code is best? Would we prefer C_I or C_{II}?

    Consider C_{II} and decode string: 0011. It could be either $1, 1, 2, 2$ or $3, 4$.

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  - Consider C_{III}. Can we decode 1100000000?

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  - Consider C_{III}. Can we decode 1100000000?

    Yes. But if we only see a prefix, such as 11, we don't know unitl we see more bits to the end.

    $1100000000 = 3, 2, 2, 2, 2$

    $11000000000 = 4, 2, 2, 2, 2$

Source Coding: Set of codes

- For $\mathcal{X} = \{1, 2, 3, 4\}$ and binary code, consider

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  - Consider C_{IV}. This code seems at least feasible (since $E\ell \geq H$). Decoding seems easy: (e.g., $111110100 = 111, 110, 10, 0 = 4, 3, 2, 1$).

Source Coding: Code types

- **Definiton** A code C is called *nonsingular*（非奇异的） if every element of $\mathcal{X}$ maps onto a difference string in $\mathcal{D}^*$, i.e.,

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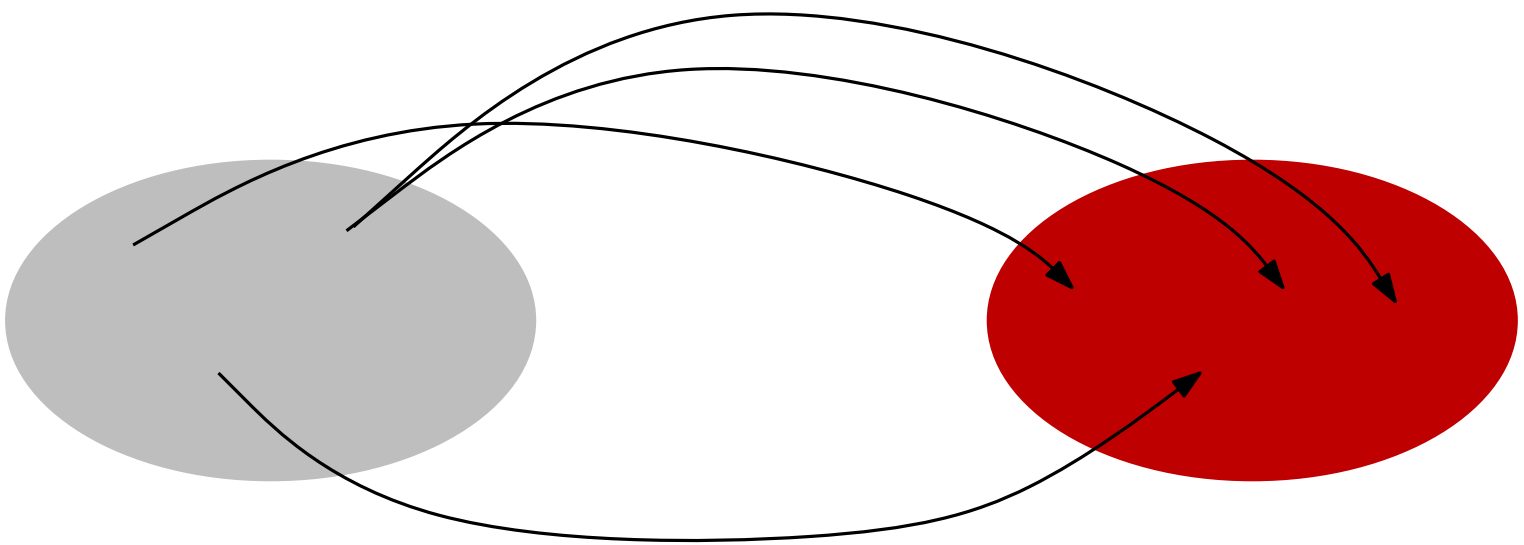

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C_I is singular.

Source Coding: Code types

- **Definiton** The *extension* of a code C: $\mathcal{X} \to \mathcal{D}^*$ is defined by

$$C(x_1 x_2 \cdots x_n) = C(x_1)C(x_2)\cdots C(x_n).$$

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C_{III} is uniquely decodable.

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$1100000000 = 3, 2, 2, 2, 2$
$11000000000 = 4, 2, 2, 2, 2$

To know the source, we have to wait until the end!

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C_{IV} is prefix.

Source Coding: Code types

- For $\mathcal{X} = \{1, 2, 3, 4\}$ and binary code, consider

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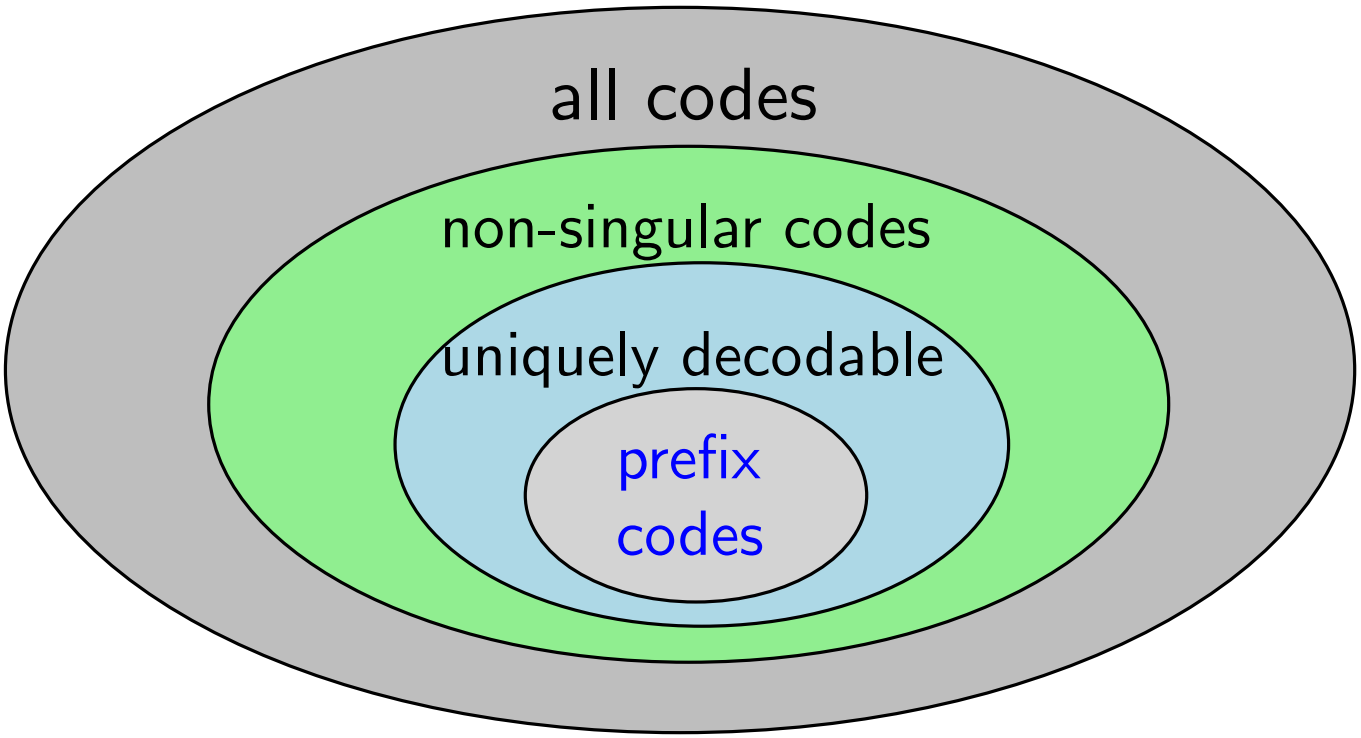
  - C_I is singular.
  - C_{II} is non-singular, but not uniquely decodable.
  - C_{III} is non-singular, uniquely decodable（唯一可解码）,
  - but NOT prefix. C_{IV} is non-singular, uniquely decodable, and prefix.

Source Coding: Classes of codes



- Goal: to find a prefix code with minimum expected length.

Kraft Inequality

- **Theorem 5.2.1** (*Kraft inequality*) For any prefix code over an alphabet of size D, the codeword lengths $\ell_1, \ell_2, \ldots, \ell_m$ must satisfy the inequality

$$\sum_i D^{-\ell_i} \leq 1.$$

Conversly, given a set of codeword lengths that satisfy this inequality, there exists a prefix code with these codeword lengths.

Kraft Inequality

- **Proof Idea.** (A small example) To prove: A prefix code with lengths $\ell_1, \ell_2, \ldots, \ell_m$, the inequality

$$\sum_i D^{-\ell_i} \leq 1 \quad \text{holds.}$$

x	$c(x)$
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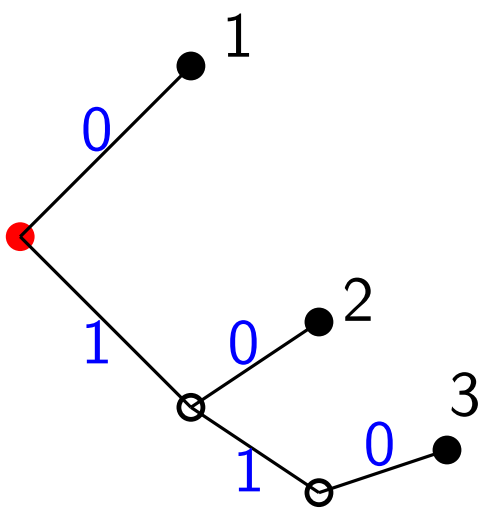

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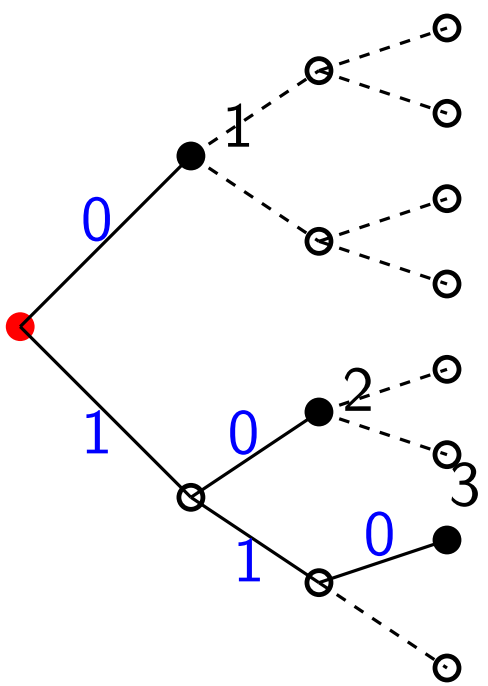
$\ell_{\max} = 3$



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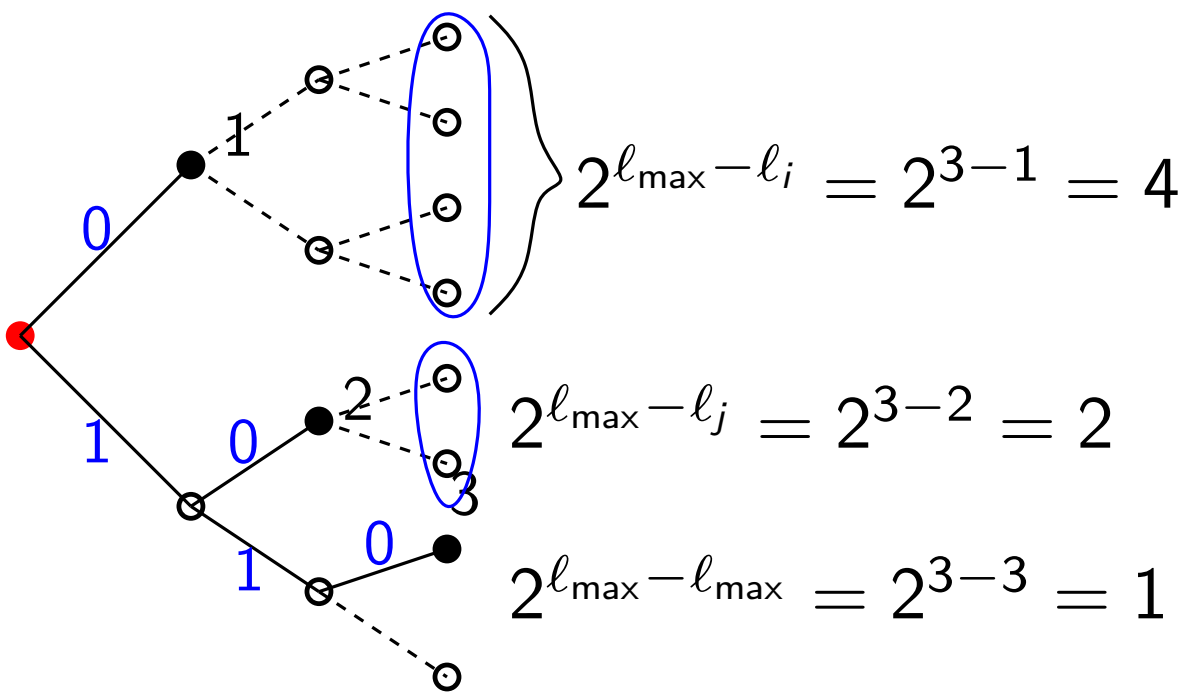

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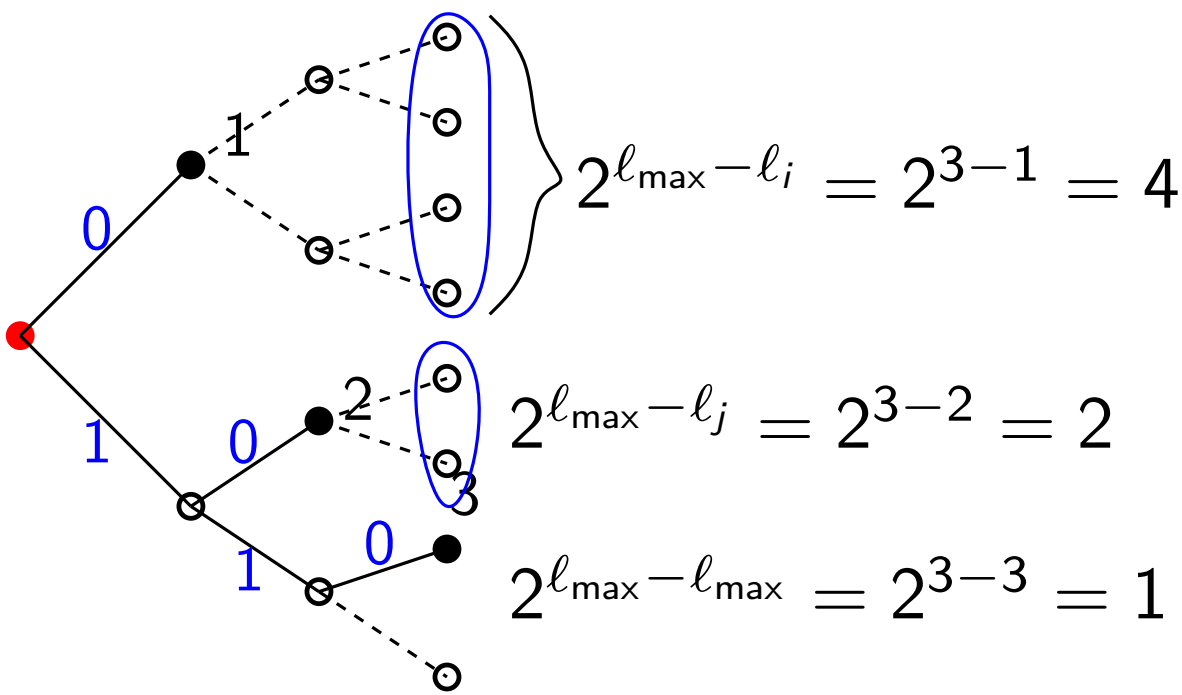


$2^{\ell_{\max}-\ell_i} = 2^{3-1} = 4$

$2^{\ell_{\max}-\ell_j} = 2^{3-2} = 2$

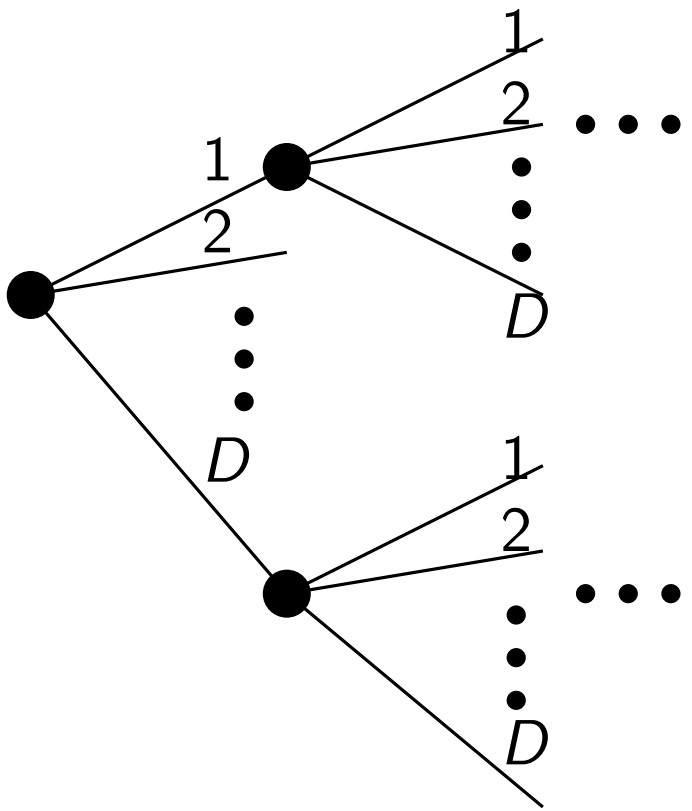
$2^{\ell_{\max}-\ell_{\max}} = 2^{3-3} = 1$

$\ell_{\max} = 3$

$\sum_i 2^{-\ell_i} \leq 1 \quad \Leftarrow \quad \sum_i 2^{\ell_{\max}-\ell_i} \leq 2^{\ell_{\max}}$

Kraft Inequality

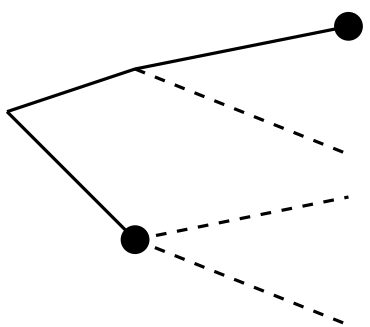
- **Proof.** (in general)
  - Represent the set of prefix codes on a D-ary tree:



  - Codewords correspond to leaves
  - Path from root to each leaf determines a codeword
  - Prefix condition: won't get to a codeword until we get to a leaf (no descendants of codewords are codewords)

Kraft Inequality

- **Proof.** (in general)
  - $\ell_{\max} = \max_i(\ell_i)$ is the length of the longest codeword.
  - We can expand the full-tree down to depth $\ell_{\max}$:



The nodes at the level $\ell_{\max}$ are either:

- codewords
- descendants of codewords
- neither

  - Consider a codeword i at depth ℓ_i in tree
  - There are $D^{\ell_{\max}-\ell_i}$ descendants in the tree at depth $\ell_{\max}$
  - Descendants of code i are disjoint from decedents of code j (prefix free condition)

Kraft Inequality

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  - All the above implies:

$$\sum_i D^{\ell_{\max}-\ell_i} \leq D^{\ell_{\max}} \quad \Rightarrow \sum_i D^{-\ell_i} \leq 1$$

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  - Conversely: given codewords lengths $\ell_1, \ell_2, \ldots, \ell_m$ satisfying Kraft inequality, try to construct a prefix code.

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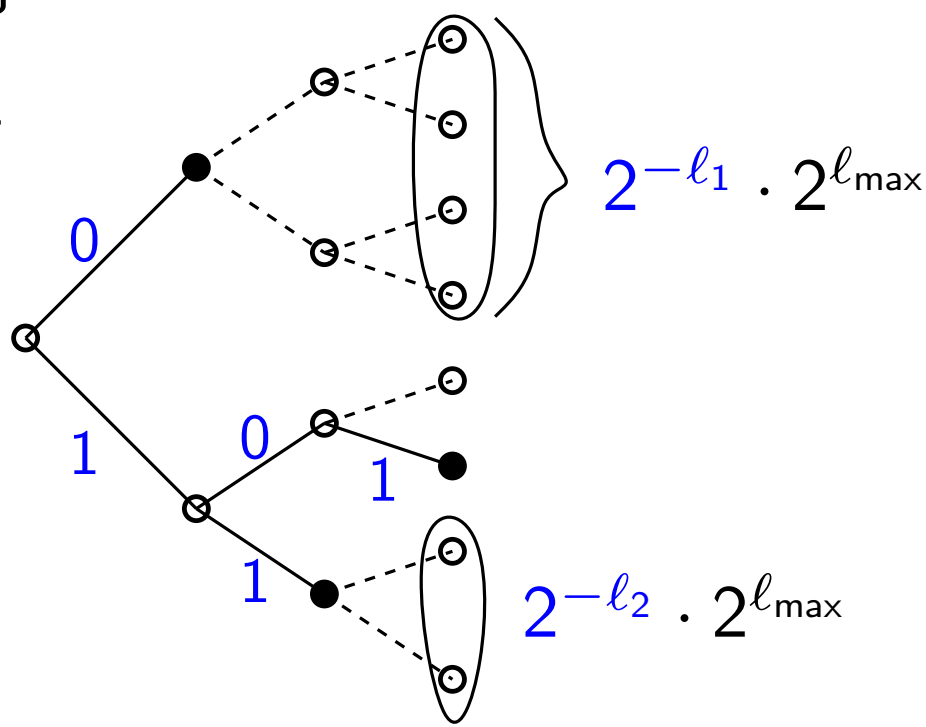

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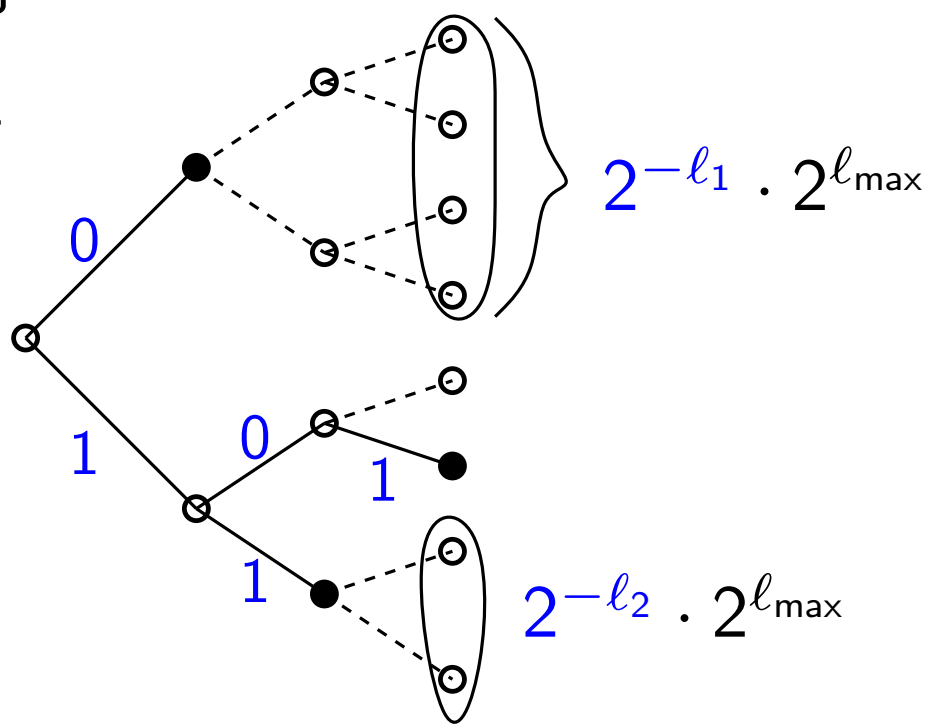
x	$c(x)$
1	0
2	11
3	101

C is prefix.



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Left as an Exersice.

Extended Kraft Inequality

- **Theorem 5.2.2** (*Extended Kraft （克拉夫特不等式） Inequality*) Kraft inequality holds also for all *countably infinite set* of codewords, i.e., the codeword lengths satisfy the extended Kraft inequality,

$$\sum_{i=1}^{\infty} D^{-\ell_i} \leq 1.$$

Conversely, given any $\ell_1, \ell_2, \ldots$ satisfying the extended Kraft inequality, we can construct a prefix code with these codeword lengths.

Extended Kraft Inequality

- **Theorem 5.2.2** (*Extended Kraft Inequality*) Kraft inequality holds also for all *countably infinite set* of codewords.

  Proof. Consider the ith codeword $y_1 y_2 \cdots y_{\ell_i}$. Let $0.y_1 y_2 \cdots y_{\ell_i}$ be the real number given by the D-ary expansion

  $$0.y_1 y_2 \cdots y_{\ell_i} = \sum_{j=1}^{\ell_i} y_j D^{-j},$$

  which corresponds to the interval

  $$\left[0.y_1 y_2 \cdots y_{\ell_i}, 0.y_1 y_2 \cdots y_{\ell_i} + \frac{1}{D^{\ell_i}}\right).$$

Extended Kraft Inequality

- **Theorem 5.2.2** (*Extended Kraft Inequality*) Kraft inequality holds also for all *countably infinite set* of codewords.

  Proof. (cont.) By the prefix condition, these intervals are disjoint in the unit interval $[0, 1]$. Thus, the sum of their lengths is ≤ 1. This proves that

  $$\sum_{i=1}^{\infty} D^{-\ell_i} \leq 1.$$

  For converse, reorder indices in increasing order and assign intervals as we walk along the unit interval.

Optimal Codes

- **Problem** To find the set of lengths $\ell_1, \ell_2, \ldots, \ell_m$ satisfying the Kraft inequality and whose expected length $L = \sum p_i \ell_i$ is minimized.

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Optimization:

minimize $L = \sum p_i \ell_i$

subject to $\sum D^{-\ell_i} \leq 1$ and ℓ_i's are integers.

Optimal Codes

- **Theorem 5.3.1** The *expected length* L of any instantaneous D-ary code for a random variable X is no less than $H_D(X)$, i.e.,

$$L \geq H_D(X),$$

with equality iff $D^{-\ell_i} = p_i$.

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Proof.

$$\begin{aligned} L - H_D(X) &= \sum p_i \ell_i - \sum p_i \log_D \frac{1}{p_i} \\ &= -\sum p_i \log_D D^{-\ell_i} + \sum p_i \log_D p_i \\ &= \sum p_i \log_D \frac{p_i}{r_i} - \log_D c \\ &= D(\mathbf{p}||\mathbf{r}) + \log_D \frac{1}{c} \\ &\geq 0, \end{aligned}$$

where $r_i = D^{-\ell_i} / \sum_j D^{-\ell_j}$ and $c = \sum D^{-\ell_i} \leq 1$.

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"=" holds if $c = 1$ and $r_i = p_i$.

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Remark $H_D(X)$ is a lower bound on the optimal code length. The equality holds iff p is D-adic.

Bound on the Optimal Code Length

- **Theorem 5.4.1** (*Shannon Codes*) Let $\ell_1^*, \ell_2^*, \ldots, \ell_m^*$ be optimal codeword lengths for a source distribution **p** and a D-ary alphabet, and let L^* be the associated expected length of an optimal code ($L^* = \sum p_i \ell_i^*$). Then

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$$H_D(X) \leq L^* < H_D(X) + 1.$$

Proof. Take $\ell_i = \lceil -\log_D p_i \rceil$. Since

$$\sum_{i \in \mathcal{X}} D^{-\ell_i} \leq \sum p_i = 1,$$

these lengths satisfy Kraft inequality and we can create a prefix code. Thus,

$$\begin{aligned} L^* &\leq \sum p_i \lceil -\log_D p_i \rceil \\ &< \sum p_i (-\log_D p_i + 1) \\ &= H_D(X) + 1. \end{aligned}$$

Bound on the Optimal Code Length

- **Theorem 5.4.2** Consider a system in which we send a sequence of n symbols from X. The symbols are assumed to be drawn i.i.d. according to $p(x)$. The *minimum expected codeword length per symbol* satisfies

$$\frac{H(X_1, X_2, \ldots, X_n)}{n} \leq L_n^* < \frac{H(X_1, X_2, \ldots, X_n)}{n} + \frac{1}{n}.$$

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Proof. First,

$$L_n = \frac{1}{n} \sum p(x_1, x_2, \ldots, x_n) \ell(x_1, x_2, \ldots, x_n) = \frac{1}{n} E\ell(X_1, X_2, \ldots, X_n).$$

We also have

$$H(X_1, X_2, \ldots, X_n) \leq E\ell(X_1, X_2, \ldots, X_n) < H(X_1, X_2, \ldots, X_n)+1.$$

Since $X_1, X_2, \ldots, X_n$ are i.i.d., $H(X_1, X_2, \ldots, X_n) = nH(X)$.

INFORMATION THEORY & CODING